Adityakeerti

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Professional Summary

Final-year AI/ML student focused on bridging the gap between complex AI models and production-ready software. Hands on experience in designing scalable backend architectures, high-throughput event processing, and robust data pipelines to support autonomous agents and computer vision applications. Passionate about building reliable, real-world systems that deliver measurable business impact.

Technical Skills

Languages: C, Python, Java, SQL

Backend & Architecture: FastAPI, REST APIs

Databases: MySQL, SQLite

AI & Processing: PyTorch, OpenCV, YOLO, LLM Orchestration, HuggingFace

Cloud & DevOps: Google Cloud (Vertex AI), Git, GitHub Actions, CI/CD

Experience

Peerprep

Software Engineering Intern — *Mar 2026 - May 2026*

- Built an automated ingestion engine using Python to standardize and extract data from unpredictable, multi-format academic documents.
- Implemented strict validation checks, reducing per-record processing time from 10 minutes to ∼30 seconds while maintaining 99.98% accuracy on internal validation datasets.

Tech Stack: Python, OpenCV, OCR, Git, GitHub

Projects

Resume Rank AI — Intelligent Candidate Ranking System GitHub | Live Demo — *May 2026 - July 2026*

- Engineered a multi-stage retrieval and ranking pipeline utilizing Matryoshka text embeddings (nomic-embed) and OpenBLAS-accelerated vector cosine similarity to shortlist the top 2,000 candidates from a pool of 100,000 in under 1 second.
- Architected a zero-shot logical validation gate using Natural Language Inference (DeBERTa-NLI) to analyze constraint alignment (e.g., work mode, relocation) and prune the shortlist to the top 500 logically compliant candidate profiles.
- Implemented an ONNX-optimized MS-MARCO Cross-Encoder to rerank the top 500 candidates down to the final top 100 best-fit candidates, utilizing parallel multiprocessing and Cumulative Distribution Function (CDF) normalization to stay well under the strict 5-minute CPU execution limit.

Tech Stack: Python, PyTorch, ONNX Runtime, DeBERTa-NLI, MS-MARCO Cross-Encoder, Sentence Transformers, NumPy, Pandas, Multiprocessing

Advanzia AutoLend - Autonomous Credit Limit Optimization GitHub | Live Demo — *Jan 2026 - Feb 2026*

- Implemented a Dueling Double DQN-based risk optimization engine inspired by existing research on dynamic credit limit optimization, combined with a Cox Proportional Hazard Model to estimate default probabilities.
- Architected a high-throughput microservices backend (Spring Boot, FastAPI) leveraging Redis caching, benchmarking throughput at 5,000+ requests/min with sub-300ms inference latency during evaluation.
- Designed an immutable double-entry ledger architecture with audit trails and built a simulation pipeline modeling 10,000+ synthetic users to pre-train and validate reinforcement learning policies.

Tech Stack: Java, Spring Boot, Python, FastAPI, PyTorch, Redis, PostgreSQL

DOC-OC - Automated Marksheet Extraction System GitHub | Live Demo — *Sep 2025 - Oct 2025*

- Engineered a multi-board extraction pipeline utilizing OpenCV, custom YOLOv8 object detection, and Hugging Face Table Transformers to automate structured data parsing for administrative workflows.
- Built a custom structure-retaining OCR engine (via LLM Whisperer) integrated with strict fraud-mitigation checks to ensure data integrity across consecutive semesters.
- Developed a React/TypeScript and async FastAPI portal backed by a secure MySQL database, driving a 30x reduction in processing time (10 mins to ∼30s per marksheet).

Tech Stack: Python, TypeScript, React, FastAPI, YOLOv8, HuggingFace, OpenCV, MySQL

Education

Graphic Era Hill University — **Dehradun**

B.Tech Computer Science — Artificial Intelligence and Machine Learning — *2023 – Present*

CGPA: 7.55/10.0

Achievements

- **1st Place, MariTHON National Hackathon** — SOF extraction pipeline built and deployed using Google Cloud Vertex AI for the Integrated Maritime Exchange.
- **Finalist, HackTheWinter National Hackathon** — AI-driven dynamic credit limit optimization system using reinforcement learning.
- **Finalist, India Innovates 2026** — Qualified for the Physical Finale of the World's Largest Civic Tech Hackathon (Top 500 Teams) | Bharat Mandapam, New Delhi, India.